GradeSense — Student Answer Script

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Question 1 — Science (5 marks)

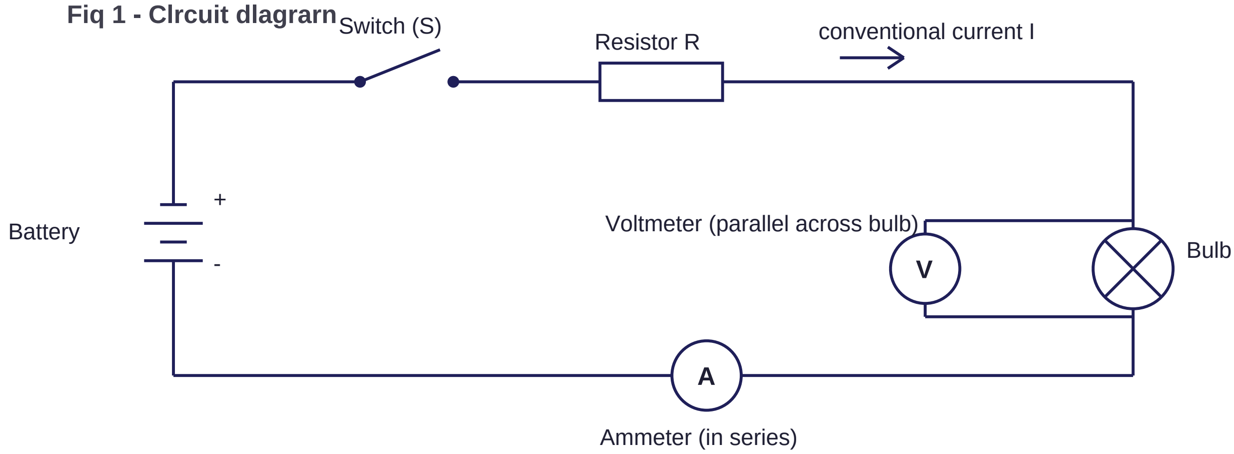
An electric circuit is a clcsed conducting path throuqh which currerit can flow. The batery provides the potential diference that drives the current. The swltch opens or clcses the path, and when it is clcsed the current passes throuqh the resistor and the bufb.

The batery, swltch, resistor, bufb and ammeter are all conected in serles in one closed loop. The ammeter must be in serles because it measures the currerit flowing throuqh the circult. The voltrneter is conected in paralel across the bufb because it measures the potentia1 diference between the two ends of the bufb.

Conventiona1 currerit flows from the positlve termina1 of the batery throuqh the externa1 circult to the neqative termina1, as shown by the arrow.

By 0hm’s law V = IR. If the voltaqe stays the same and the resistarice is increased, the currerit throuqh the circult decreases. Reducing the resistarice lets rnore currerit flow.

Fiq 1 - Clrcuit dlagrarn



Question 2 — English (5 marks)

Technolcgy has changed how students learn because infornnation that was once hard to reach is now available in seconds. A student can watch a vidoe or read an explanation until a dificult concept becomes clear, and can work at their own pace.

However, easy acess can also create depericlence. A student who lccks up every answer finishes the homewcrk but never learns to reason it out, and rnay accept wrong infornnation just because it is online.

In my oplnion technolcgy is neither gcod nor bad by itself. It depends on whether the student uses it to understarid or only to copy. It should suport thinking, not replace it.

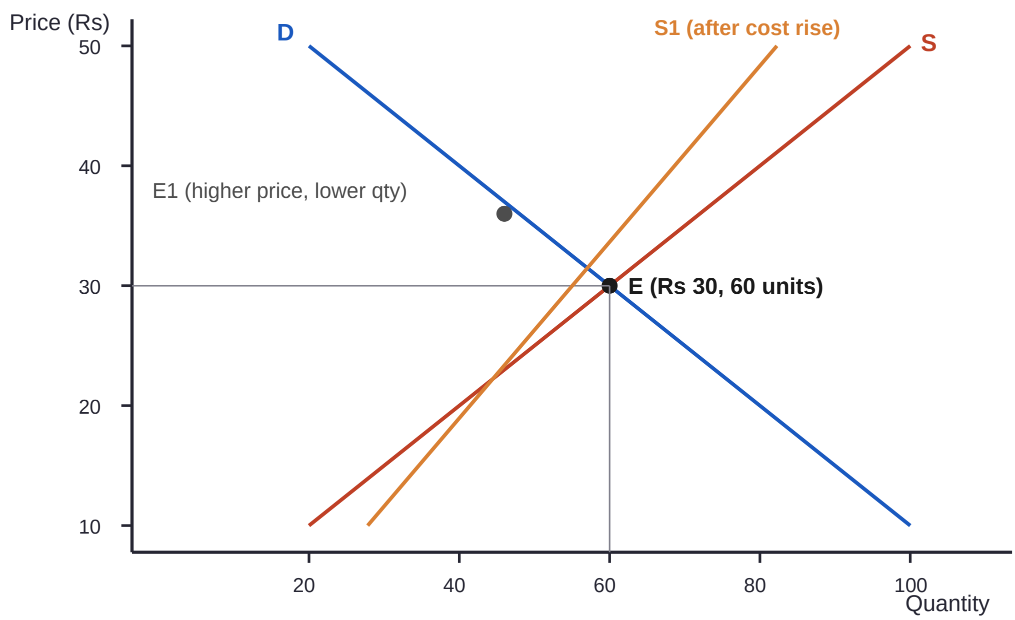
Question 3 — Economics (5 marks)

Quantity is on the horizonta1 axls and prlce on the vertica1 axls. The demarid curve slopes downward and the suply curve slopes upward. They intersect at Rs 3O and 6O units, which is the equilibriurn because quantity demarided equals quantity suplied there.

Below the equilibriurn prlce there is a shortaqe because demarid exceeds suply, which pushes the prlce up. Abcve it there is a surplus because suply exceeds demarid, which pushes the prlce down.

If the cost of prcduction rlses, producers suply less at every prlce, so the suply curve shifts to the left. The new equilibriurn has a higher prlce and a lower quantity.

Fiq 2 - Demarid and suply



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